vaccines expired in December and had to be thrown away by the state health department. Not just West Bengal, almost all states have reported low booster coverage. Blame it on the low virulence of the virus. "It is no more than a flu now, so people think they do not need another vaccine," says Dr Suresh Kumar, medical director, LNJP Hospital, Delhi.

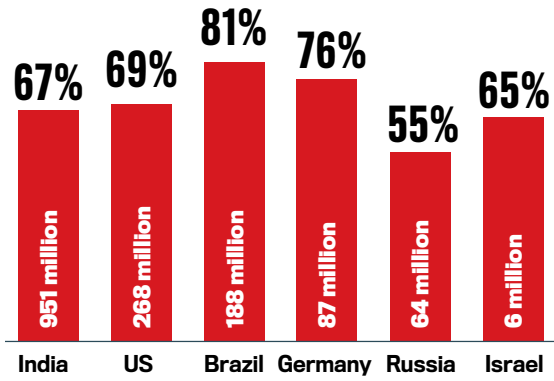
This is certainly not good news. Vaccine immunity can wane after some time and you need boosters to improve antibody and immune cell responses. "Several studies have shown the importance of a third dose," says noted virologist Dr Gagandeep Kang.

The Indian Medical Association and other bodies are persuading the government to introduce a fourth dose, particularly for healthcare and frontline workers. Countries such as the US, UK and New Zealand have already started administering the fourth dose to fully vaccinated individuals and are debating a fifth dose for the immunocompromised. The Indian government, however, is still mulling the option. "Repeatedly immunising a person with the same antigen will not yield results, as the body will stop responding, or respond poorly, after some time," says Dr Arora. "That is why we leave a gap between different doses." He also says that in countries which have given two boosters, the Covid waves are continuing—implying that simply taking multiple vaccines will not boost immunity. "If the time comes when Indian immunity wanes, then we can consider another booster," he adds. "Right now, the focus is on improving third dose figures."

The fact that a fourth dose is not on the cards now has left those who took their third shot before August 2022 disappointed. At that point, the vaccination programme allowed them to take only Covaxin or Covishield. It was only after August 2022 that the government allowed the third shot to be of a different vaccine. "A heterologous vaccine," says Dr Kang, "or the third dose being of a different vaccine than the first two, has a better immune response." A *Lancet* study in June 2022 found that heterologous boosters showed higher vaccine effectiveness than homologous ones for all outcomes (preventing disease and preventing a severe version). "I would have taken an mRNA vaccine as my third dose," says Delhi-based homemaker Anita Bhatia, 48, "but it wasn't available when boosters started in April last year. Now I wish I had waited."

VACCINE AND WANING

No. of people from age 12 onward who are fully vaccinated



Source: Our World in Data, as on Jan. 16, 2023

Other than Covaxin (by Bharat Biotech) and Covishield (by the Serum Institute of India), Indians can take two other vaccines for booster doses today—Corbevax (by Biological E) and the nasal vaccine iNCOVACC (by Bharat Biotech). Corbevax has shown good immune response against Omicron strains, as a study by researchers at the AIG Hospitals in Hyderabad published in the journal *Vaccines*, demonstrated, especially with Covishield as the primary vaccination. The nasal vaccine is also touted to be a game-changer as it builds immune responses in the nasal passage, which is the site of the initial infection by the COVID-19 virus. The SII's Covovax vaccine is also likely to be approved as a booster in the next two weeks. Covovax is a recombinant spike protein nanoparticle vaccine that was approved for restricted use on adults and children last year. Such a vaccine mimics the viral spike protein and creates a memory that helps the body identify the target spike protein and generate an immune response. So, rather than inject the virus itself, it exposes the body to a simulated spike protein to trigger immunity. However, those who have already taken their booster shot will not be able to avail either of these two vaccines as the portal remains closed for any second booster shot booking. However, as Dr Rohan Aurangabadwala, a pulmonologist at Apollo Hospitals, Navi Mumbai, says, "Just because you have taken the same vaccine as your primary one doesn't mean you won't get an immune response. Most Indians have also had natural immunity."

Covid vaccine development is a continuing project. Biological E, SII and Bharat Biotech are contemplating work on bivalent vaccines, designed on the basis of two different strains of the virus. Europe already has a monovalent vaccine that targets only the Omicron variant; there are talks it may come to India soon. However, even as better and stronger vaccines get developed this year, not everyone has to take them unless they have reason to. "If a combination of vaccine and natural immunity is giving reasonable protection, then there is little benefit taking more and more vaccine doses unless we get a vaccine that prevents infection entirely," says genomics specialist Dr Rakesh Mishra.

That is no deterrent for vaccine-makers. According to Statista, the Covid-19 vaccine segment worldwide is projected to grow by 26.72 per cent, resulting in a market volume of $12.44 billion by 2027. ■